MIL-A-23088A(SHIPS)
24 June 1963
SUPERSEDING
MIL-A-23088(SHIPS)
14 November 1961

MILITARY SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY AM-3198( )/URC

1. SCOPE

1.1 This specification covers the amplifier, audio frequency AM-3198( )/URC. This amplifier is normally for use within radio sets AN/URC-32( ), radio transmitter AN/URT-18 or Collins Radio Company Model KWT-6( ), or equivalent, for applications using A3j mode of emission and external audio inputs. This amplifier limits voice peaks at the output and provides compression and clipping features to increase the average audio level of a given signal. This replaceable electronic assembly is normally used in lieu of the originally designed upper and lower sideband line amplifier units.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-C-5541 - Chemical Films for Aluminum and Aluminum Alloys.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.

STANDARD

MILITARY
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. - A preproduction sample shall be tested as specified in 4.3 (see 6.2).

3.2 Material and parts. - Material and parts shall be in accordance with MIL-E-16400, except as otherwise specified herein.

3.3 Design. - The design of the equipment shall be the same as the Government-furnished equipment (GFE), unless specifically approved by the bureau or agency concerned.

3.4 Characteristics. -

3.4.1 Audio input. - The unit shall be capable of accepting normal audio signals in the range of 200 to 6000 cycles per second (c. p. s.).

3.4.2 Type of service. - The unit shall be capable of continuous, unattended operation.

3.4.3 Size. - The overall dimensions of the complete unit shall not exceed 3-1/4 inches in height, 3 inches in length, and 2 inches in width.

3.4.4 Type of mounting. - The unit shall be held to the mounting panel with two 8-32 captive type screws, mounted on diagonal corners.

FSC 5820

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3.4.5 Connector.- The units connector shall be a six-contact printed circuit type to mate with Winchester Electronics type K6S or Collins Radio Company type 372-1568-00 receptacle, or equivalent, presently in the radio set AN/URC-32.

3.4.5.1 The terminal connections for this connector shall be as follows:

Terminal	Description
A	B + (28 volts direct current (vdc) nominal))
B	No connection
C	Audio input (150 ohms)
D	Circuit return (ground)
E	Circuit return (ground)
F	Audio output (600 ohms)

3.4.6 Temperature operating range.- The temperature operating range shall be -40° to +50°C.

3.4.7 Ambient humidity range.- The ambient humidity range shall be up to 95 percent relative humidity at +50°C.

3.4.8 Shock.- The unit shall be capable of two shocks of 35 gravity (g), for a duration of 11 ± 1 milliseconds in a normal mounting plane. The unit shall be capable of two shocks at 20 g in each of four horizontal directions. The unit shall be nonoperating during the test and performance shall be maintained after tests in each plane.

3.4.9 Vibration.- The unit shall be capable of meeting the requirements of type I, 5 to 33 c.p.s. (nonoperating) of MIL-STD-167. Performance shall be maintained after vibration in each plane. Total running time shall be 90 minutes in each of three mutually perpendicular planes.

3.4.10 Power source.- Primary power input shall be 22 to 30 v.d.c. or as would be supplied by a dual section choke input filter that is fed from a 26-vdc, unfiltered full wave rectifier circuit (120-c.p.s. ripple). Ripple shall not exceed 100 millivolts.

3.4.11 Power demand.- The power demand shall not exceed 80 milliampere direct current (d.c.) (at 24 volts).

3.4.12 Input impedance.- The amplifier units input impedance shall be 150 ohms, ±25 percent, unbalanced transformer input.

3.4.13 Normal input signal level.- The normal signal level shall be -37.8 dbm.

3.4.14 Dynamic input range.- Compression characteristics shall be maintained from -37.8 dbm to -4 dbm input.

3.4.15 Output impedance.- The units output impedance shall be 600 ohms ±25 percent, unbalanced transformer type.

3.4.16 Normal output signal level.- The normal output signal level shall be +8.4 dbm with a capability of + 13 dbm maximum for normal input specified in 3.4.13.

3.4.17 Frequency response.- The frequency response shall be 300 to 3000 c.p.s. ±2 decibels (db) at normal output signal level, +6 db for conditions of 3.4.6 through 3.4.9.

3.4.18 Total harmonic distortion.- The total harmonic distortion shall be not more than 6 percent.

3.4.19 Noise.- Noise shall be at least 40 db below normal output of 3.4.16.

3.4.20 Compression characteristics.-

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3.4.20.1 Threshold. - Compression threshold shall occur within ±3 db of normal output of 3.4.16, when normal input of 3.4.13 is applied.

3.4.20.2 Rise in output. - The output shall be maintained within ±3 db of normal output over the dynamic input range of 3.4.16.

3.4.21 Clipping characteristics. - Symmetrical peak clipping shall be provided, and shall occur after signal compression. Clipping threshold shall be set 0.5 to 4 db above maximum output specified in 3.4.20.2.

3.4.22 Time constants. -

3.4.22.1 Attack. - Attack time shall be defined as total time required for the output to change from the normal value (see 3.4.16) and to first reach 90 percent of its finalized value, when the input is stepped from -37.8 dbm to -27.8 dbm. This time shall be 10 to 50 milliseconds.

3.4.22.2 Release. - Release time shall be 1 to 2 seconds.

3.4.23 Controls. - An adjustable gain control shall be provided within the unit and shall provide output adjustments from +13 dbm to -10 dbm with normal input specified in 3.4.13.

3.5 Finish. - All aluminum surfaces shall be treated in accordance with MIL-C-5541.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection. - Examination and tests to be performed on the equipment shall be in accordance with MIL-E-16400 unless otherwise specified in the contract or order. Inspection records of the examination and tests shall be kept complete for a period not less than 2 years after delivery of equipment. Test records shall be shipped with the tested equipment when the equipment is offered for delivery. Examination and tests shall consist of the following:

(a) Preproduction inspection, if specified in the contract or order (see 6.1).
(b) Quality conformance inspection, including production and production control inspection.

4.3 Preproduction inspection. - Preproduction (production type) models, when required by the contract or order (see 6.1) shall be inspected to determine conformance with this specification. Preproduction equipment offered for approval shall, prior to delivery, be completely inspected in accordance with the preproduction inspection requirements of MIL-E-16400, except that tests for water cooling and enclosure are not required. Any additional tests necessary to determine compliance with this specification shall also be conducted. Preproduction approval by the Government is required prior to delivery of production equipment. A report of the preproduction inspection shall be submitted to the cognizant Government agency immediately following completion of tests.

4.4 Production inspection. - Production inspection shall be conducted in accordance with MIL-E-16400.

4.4.1 Production control inspection. - Production control inspection tests shall be conducted on one out of every 15 equipments that have successfully passed production inspection or one equipment from a month's production, whichever is the minimum. The selection of equipments for production control testing shall be made on a month-to-month basis. Tests shall be conducted on current production equipments and not on equipments produced during previous months. Production control inspection shall consist of the tests specified in MIL-E-16400, except that the weld test and water cooling test will not be required.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

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5.1.1 Amplifier, Audio Frequency AM-3198( )/URC.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method or packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation and packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing, and marking.- The amplifier, and technical publications, shall be preserved and packaged by level A or C, packed by level A, B, or C, and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether preproduction model is required (see 4.3).
(c) Preservation and packaging, packing, and marking requirements if other than specified in 5.1 (see 5.2).

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has previously been procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - Ships
(Project 5820-N343Sh)

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-1863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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